OPERATING SYSTEMS

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File Management

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OPERATING SYSTEMS

RAID

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Slides Credits for all PPTs of this course

- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources and persons:

1. Slides of Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - 9th edition 2013 and some slides from 10th edition 2018
2. Some conceptual text and diagram from Operating Systems - Internals and Design Principles, William Stallings, 9th edition 2018
3. Some presentation transcripts from A. Frank – P. Weisberg
4. Some conceptual text from Operating Systems: Three Easy Pieces, Remzi Arpaci-Dusseau, Andrea Arpaci Dusseau

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RAID Basics

- RAID is a Disk Organization technique
  - Addresses performance and reliability issues
  - Multiple disk drives provide (or improve) reliability via **redundancy**
- Many systems today need to store many terabytes of data
- Don't want to use single, large disk
  - too expensive
  - failures could be catastrophic
- Would prefer to use many smaller disks
- **Redundant Array of Independent (inexpensive) Disks**

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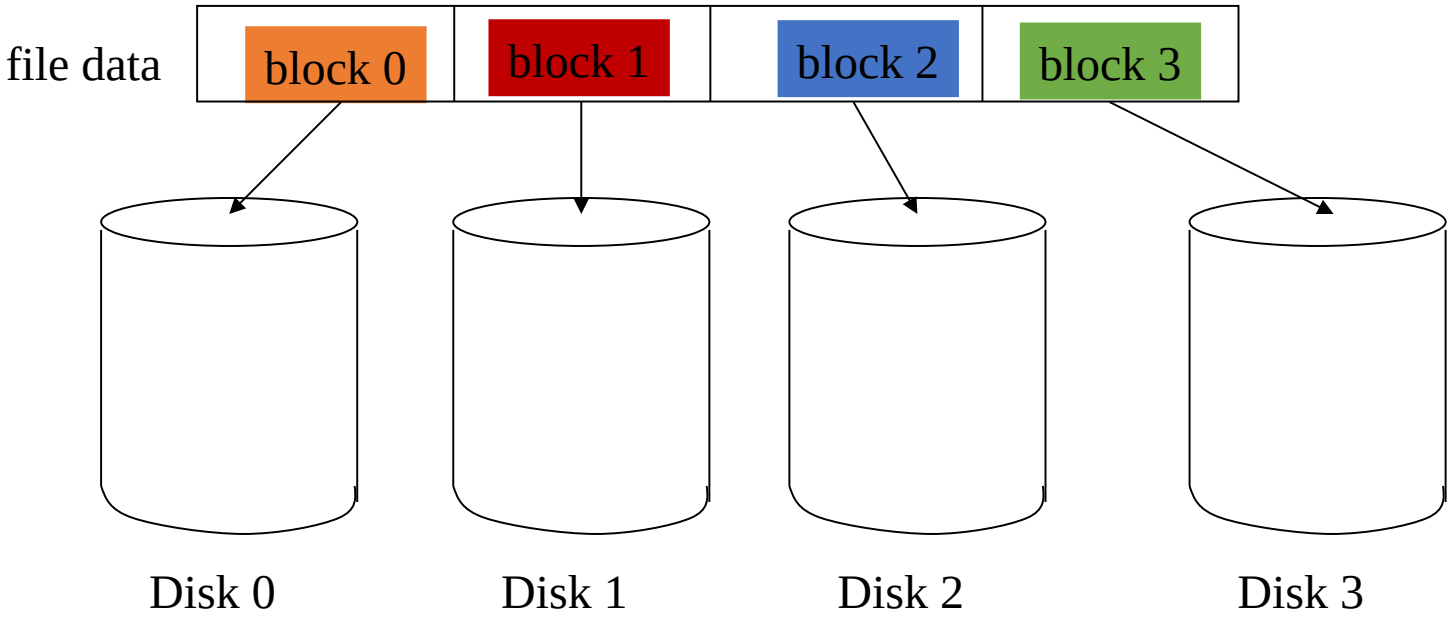
RAID Basics (Cont.)



- Basic idea is to connect multiple disks together to provide
  - large storage capacity
  - faster access to reading data
  - redundant data
- Many different levels of RAID systems
  - differing levels of redundancy, error checking, capacity, and cost

Striping

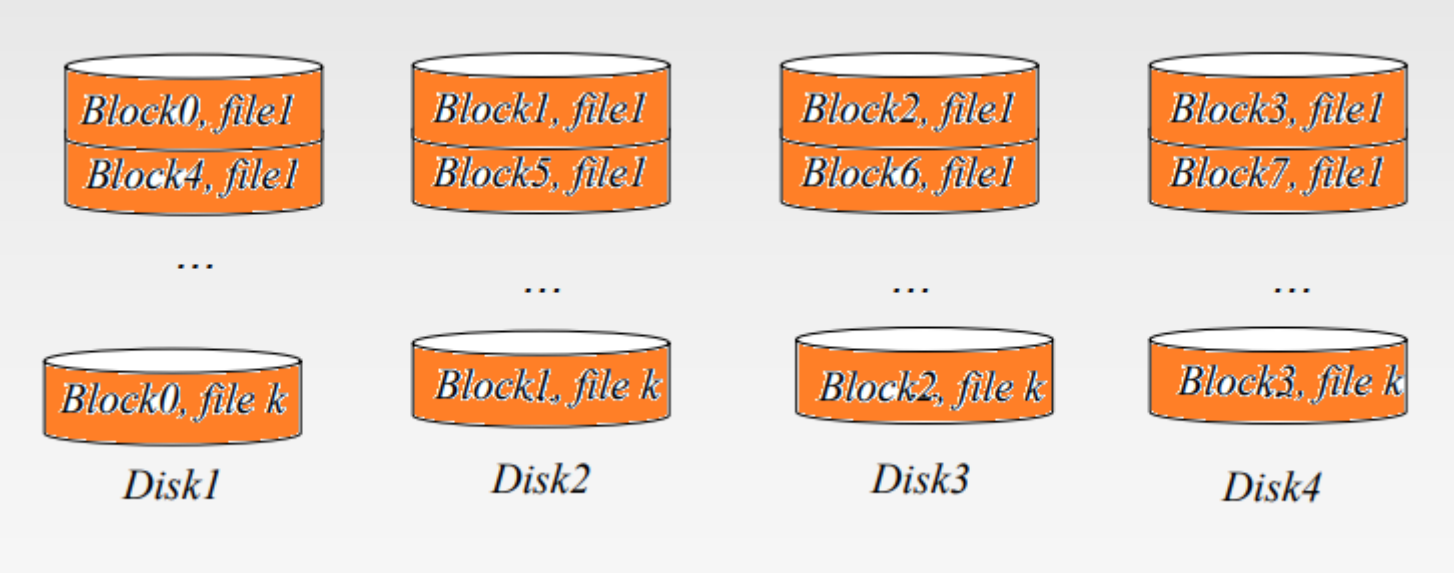
- Take file data and map it to different disks
- Allows for reading data in parallel
- Transfer rate is improved by striping data across the disks
  - **Bit-level striping and block-level striping (most common)**



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Block-level Striping

- Requests for different blocks can run in parallel if the blocks reside on different disks
- A request for a long sequence of blocks can utilize all disks in parallel



Mirroring



- Keep two copies of data on two separate disks
- Gives good error recovery
  - if some data is lost, get it from the other source
- Expensive
  - requires twice as many disks
- Write performance can be slow
  - have to write data to two different spots
- Read performance is enhanced
  - can read data from file in parallel

RAID Structure



- **Mean time to failure**
  - If MTTF of a single disk is 100,000 hours, MTTF of some disk in an array of 100 disks = 100000/100 = 1000 hours or 41.66 days
- **Mean time to repair –** time taken to replace a failed disk and to restore the data on it
- **Mean time to data loss** based on above factors
  - If mirrored disks fail independently (i.e., not related to power failures and natural disasters), consider disk with MTTF of 100,000 hours and MTTR is 10 hours

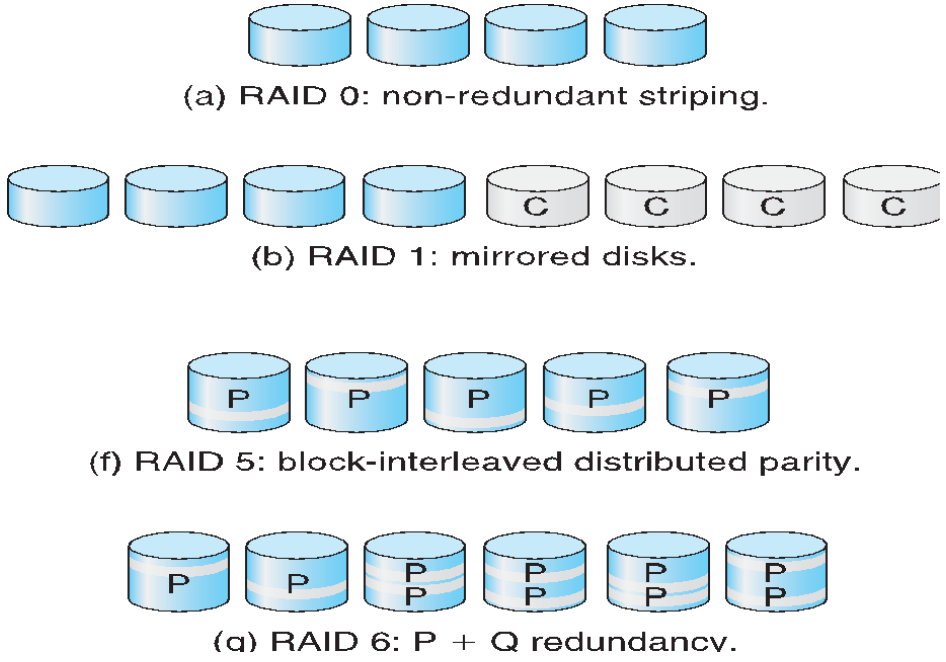
RAID (Cont.)



- Several improvements in disk-use techniques involve the use of multiple disks working cooperatively
- Disk **striping** uses a group of disks as one storage unit
- RAID is arranged into six different levels
- RAID schemes improve performance and improve the reliability of the storage system by storing redundant data
  - **Mirroring** or **shadowing** (**RAID 1**) keeps duplicate of each disk
  - Striped mirrors (**RAID 1+0**) or mirrored stripes (**RAID 0+1**) provides high performance and high reliability
  - **Block interleaved parity** (**RAID 4, 5, 6**) uses much less redundancy

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RAID Levels



- P indicates error-correcting bits
- C indicates a second copy of the data
- P + Q redundancy scheme uses 2 parity values P and Q

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RAID Levels 0 and 1

- **RAID level 0** refers to disk arrays with striping at the level of blocks
  - No redundancy (such as mirroring or parity bits)
  - lots of disks means low Mean Time To Failure (MTTF)



(a) RAID 0: non-redundant striping.

- **RAID level 1** refers to disk mirroring.
  - A complete file is stored on a single disk
  - A second disk contains an exact copy of the file
  - Provides complete redundancy of data
  - file data can be read in parallel
  - must write the data out twice
  - Most expensive RAID implementation
  - requires twice as much storage space



(b) RAID 1: mirrored disks.

RAID Level-5

- **RAID level 5** stripes file data and checks data over all the disks
  - no longer a single check disk
- Consists of block-level striping with distributed parity.
  - Unlike RAID 4, parity information is distributed among the drives
- Drastically improves the performance of multiple writes
  - they can now be done in parallel
- read and write performance close to that of RAID Level-1



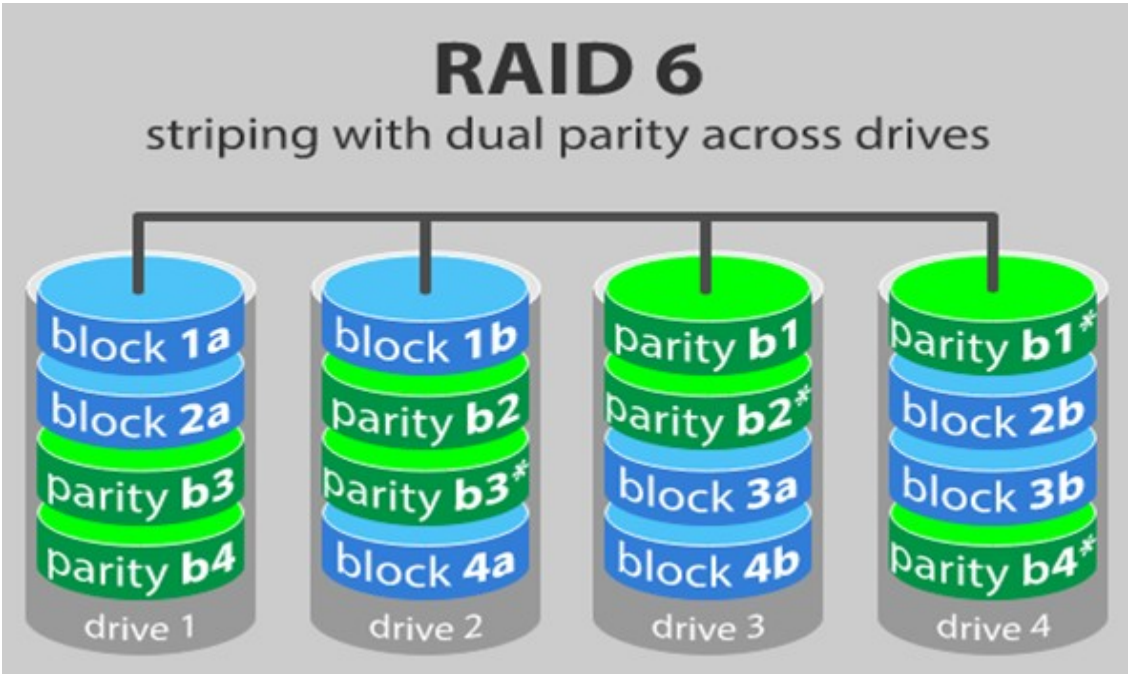
(f) RAID 5: block-interleaved distributed parity.

RAID Level-6

- RAID 6 is like RAID 5, but the parity data are written to two drives.
- Write data transactions are slower than RAID 5 due to the additional parity data that have to be calculated.

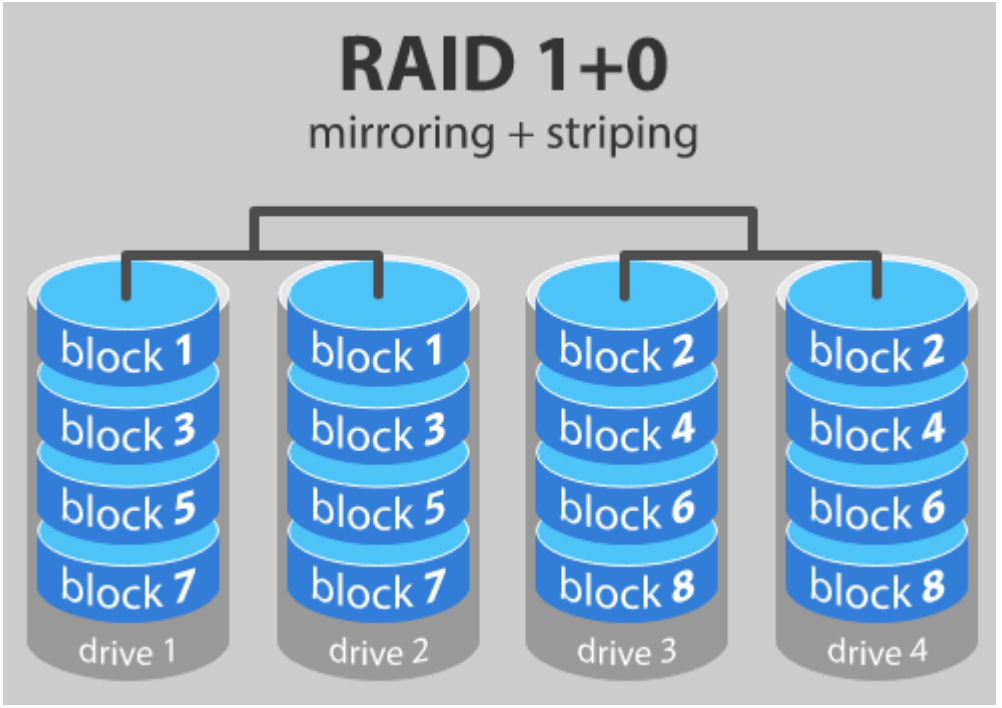


(g) RAID 6: P + Q redundancy.



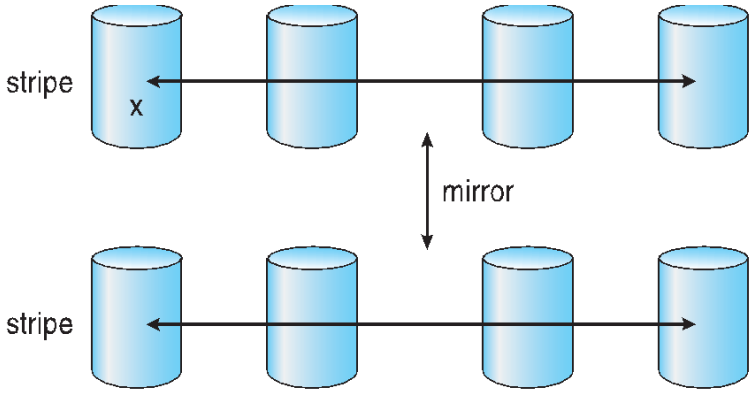
RAID level 10 - combining RAID 1 and RAID 0

- Provides security by mirroring all data on secondary drives while using striping across each set of drives to speed up data transfers.
- Rebuild time is very fast
- Half of the storage capacity goes to mirroring
  - so compared to large RAID 5 or RAID 6 arrays, this is an expensive way to have redundancy.

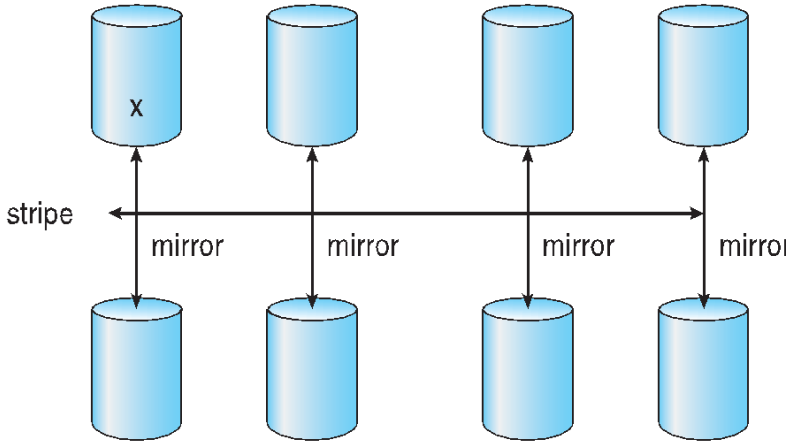


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RAID (0 + 1) and (1 + 0)



a) RAID 0 + 1 with a single disk failure.



b) RAID 1 + 0 with a single disk failure.

Disks are striped and then the stripe is mirrored to another, equivalent stripe. If a single disk fails, an entire stripe is inaccessible.

Disks are mirrored in pairs and then the resulting mirrored pairs are striped. This is better than 0+1 when a single disk fails

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THANK YOU

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